WORKSHEET

Chapter 7: Integration by substitution

1 Find the following indefinite integrals.

a $\int (3x - 1)^7 dx$

b $\int \sqrt{7x + 9}\, dx$

c $\int 3x\sqrt{x^2 - 9}\, dx$

d $\int 2x\,(x^2 + 5x)^7 dx$

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2 Find the following indefinite integrals.

a $\int 4\cos(3x)\,dx$

b $\int \frac{\sin(\sqrt{x})}{\sqrt{x}}\,dx$

c $\int \cos(x)\sin(2x)\,dx$

d $\int x\sin(2x^2)\,dx$

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3 Find the following indefinite integrals.

a $\int e^{5x+2} dx$

b $\int xe^{x^2} dx$

c $\int 3xe^{-x^2} dx$

d $\int 5x^2 e^{2x^3} dx$

4 Find the following indefinite integrals.

a $\int (3x-1)^7 dx$

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b $\int \sqrt{7x + 9}\, dx$

c $\int 3x\sqrt{x^2 - 9}\, dx$

d $\int (2x + 5)(x^2 + 5x)^7 dx$

5 Find the following indefinite integrals.

a $\int 2x\sqrt{3 + x^2}\, dx$

b $\int x\sqrt{x + 2}\, dx$

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c $\int x(x^2 + 7)^{15}\, dx$

d $\int x\sqrt{x - 3}\, dx$

6 Find the following indefinite integrals.

a $\int \cos^2(x)\sin^3(x)dx$

b $\int \sin(x)\cos^4(x)dx$

c $\int \sec^2(x)\tan^6(x)dx$

d $\int \cos(x)\sin^5(x)dx$

7 **a** Show that an anti-derivative of the expression $\cos^2(x)$ is $\frac{\sin(2x)}{4} + \frac{x}{2} + c$.

b Evaluate $\int_0^{\frac{1}{2}} \cos^2(x)dx$.

8 **a** Show that an anti-derivative of the expression $\sin(2x)\cos(2x)$ is $\frac{\sin(4x)}{32} + \frac{\sin(2x)}{4} + \frac{3}{8} + c$.

b Calculate $\int_0^1 \sin(2x)\cos(2x)dx$.

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9 **a** Show that an anti-derivative of the expression $\sin^3(x)\cos^2(x)$ is $\frac{\cos^5(x)}{5} - \frac{\cos^3(x)}{3} + c$.

b Evaluate $\int_0^{\frac{1}{3}} \sin^3(x)\cos^2(x)dx$.

10 Evaluate $\int_0^1 \cos^4(x)dx$.

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Answers

1 **a** $\frac{(3x-1)^8}{24}+c$

b $\frac{2\sqrt{(7x+9)^3}}{21}+c$

c $\sqrt{(x^2-9)^3}+c$

d $\frac{(x^2+5)^8}{8}+c$

2 **a** $\frac{4\sin(3x)}{3}+c$

b $-2\cos(\sqrt{x})+c$

c $\frac{-2\cos^3(x)}{3}+c$

d $\frac{\cos(2x^2)}{4}+c$

3 **a** $\frac{e^{5x+2}}{5}+c$

b $\frac{e^{x^2}}{2}+c$

c $\frac{-3e^{-x^2}}{2}+c$

d $\frac{5e^{2x^3}}{6}+c$

4 **a** $\frac{(3x-1)^8}{24}+c$

b $\frac{2\sqrt{(7x+9)^3}}{21}+c$

c $\sqrt{(x^2-9)^3}+c$

d $\frac{(x^2+5x)^8}{8}+c$

5 **a** $\frac{2\sqrt{(3+x^2)^3}}{3}+c$

b $\frac{2\sqrt{(x+2)^5}}{5}-\frac{2\sqrt{(x+2)^3}}{3}+c$

c $\frac{(x^2+7)^{16}}{32}+c$

d $\frac{2\sqrt{(x-3)^5}}{5}+2\sqrt{(x-3)^3}+c$

6 **a** $\frac{1}{5}\cos^5(x)-\frac{1}{3}\cos^3(x)+c$

b $\frac{-\cos^5(x)}{5}+c$

c $\frac{\tan^7(x)}{4}+c$

d $\frac{\sin^6(x)}{6}+c$

7 **a** Use the substitution: $\cos(2x)=2\cos^2(x)-1$.

b 0.460

8 **a** Use the substitution: $\cos^2(x)=\frac{\cos(2x)+1}{2}$.

b 0.267

9 **a** Use the substitution: $\cos^2(x)=\frac{\cos(2x)+1}{2}$.

b 0.002 67

10 0.579.

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